

Questions with Answer Keys

MathonGo

Q1 (20 July 2021 Shift 1)

The amplitude of wave disturbance propagating in the positive x -direction is given by $y = \frac{1}{(1+x)^2}$ at time $t = 0$ and $y = \frac{1}{1+(x-2)^2}$ at $t = 1$ s, where x and y are in metres. The shape of wave does not change during the propagation. The velocity of the wave will be ____ m/s

Q2 (20 July 2021 Shift 1)

The frequency of a car horn encountered a change from 400 Hz to 500 Hz. When the car approaches a vertical wall. If the speed of sound is 330 m/s. Then the speed of car is ____ km/h.

Q3 (20 July 2021 Shift 2)

With what speed should a galaxy move outward with respect to earth so that the sodium-D line at wavelength $5890\text{\AA}$... is observed at $5896\text{\AA}$...?

- (1) 306 km/sec
- (2) 322 km/sec
- (3) 296 km/sec
- (4) 336 km/sec

Answer Key

Q1 (2)

Q2 (132)

Q3 (1)

Hints and Solutions

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Q1

$$\text{At } t = 0, y = \frac{1}{1+x^2}$$

$$\text{At time } t = t, y = \frac{1}{1+(x-vt)^2}$$

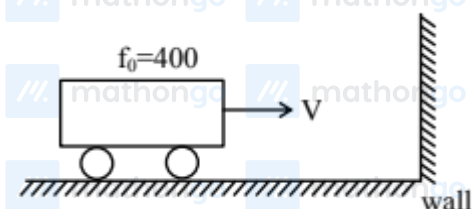
$$\text{At } t = 1, y = \frac{1}{1+(x-v)^2} \dots (i)$$

$$\text{At } t = 1, y = \frac{1}{1+(x-2)^2} \dots (ii)$$

Comparing (i) & (ii)

$$v = 2 \text{ m/s}$$

Q2



Wall as an observer

Frequency received by wall

$$f_1 = f_0 \left(\frac{C}{C-V} \right)$$

Again wall as a source

Frequency received by observer on car

$$f_2 = f_1 \left(\frac{C+V}{C} \right)$$

$$f_2 = f_0 \left(\frac{C+V}{C-V} \right)$$

$$500 = 400 \left(\frac{C+V}{C-V} \right)$$

$$\frac{5}{4} = \frac{C+V}{C-V}$$

$$C = 9V$$

$$V = \frac{C}{9} = \frac{330}{9} \text{ m/s}$$

$$V = \frac{330}{9} \times \frac{18}{5} = 132 \text{ km/h}$$

Q3

Hints and Solutions

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$$f = f_0 \sqrt{\frac{1+\beta}{1-\beta}} \quad \beta = \frac{v}{c}$$

$$\frac{f}{f_0} \sqrt{\frac{1+\beta}{1-\beta}}$$

$$\left(1 + \frac{\Delta f}{f_0}\right)^2 = (1 + \beta)(1 - \beta)^{-1}$$

β is small compared to 1

$$\left(1 + \frac{2\Delta f}{f_0}\right) = (1 + 2\beta)$$

$$\beta = \frac{\Delta f}{f_0} = \frac{v}{c}$$

$$v = 6 \times \frac{c}{5890} = 305.6 \text{ km/s}$$